

KioskControl UG: Master Technical Specification

Version 1.1 — Core Architectural & Compliance Blueprint

1. Regulatory Architecture & Compliance Framework

To satisfy Uganda's **Data Protection and Privacy Act 2019**, overseen by the Personal Data Protection Office (PDPO) under the National Information Technology Authority (NITA-U), KioskControl enforces privacy-by-design at the terminal device level.

ONBOARDING CONSENT WORKFLOW
1. Launch App
2. Display "Granular Data Disclosure Agreement"
3. Explicit Opt-In: [Click "I Accept Terms & Scan"]
4. Local SQLite logs Timestamp & Hardware Hash
5. Register local SQLite UUID with PDPO registry

1.1 First-Launch Consent Script (The Legal Shield)

Upon initial launch, the application blocks the home screen and renders a scrollable, plain-language legal notice. This is not a generic Terms of Service agreement; it is a legally binding disclosure designed to protect the platform under Ugandan law.

```
=====
                        DATA PRIVACY DISCLOSURE
=====

To automate your daily transaction ledger and protect your business from fraud,
KioskControl requests permission to read transaction notifications on this device

BY TAPPING "I ACCEPT & ACTIVATE":
1. You grant KioskControl explicit authorization to parse native transaction
   notifications sent by mobile money networks (MTN Uganda, Airtel Uganda)
   and banking agents.
2. Your business records are stored securely ONLY on this physical phone's
   encrypted memory. We DO NOT send your transaction data to external servers
   without your direct permission.
3. You retain the right to erase all captured data instantly at any time via
   the "Wipe Data & Reset" tool in your settings panel.
4. Data capture is performed entirely locally and offline, minimizing risk
   and protecting your financial privacy.
=====
                        [ I DECLINE ]           [ I ACCEPT & ACTIVATE ]
```

1.2 PDPO Registration & Compliance Strategy

- **Official Registration:** We will register KioskControl with the Personal Data Protection Office (PDPO) as a "Data Controller" and "Data Processor" using the standard Form 1 interface.
- **The Compliance Officer Role:** The owner profile inside the app functions as the designated data supervisor, validating that all transaction histories and staff records meet local privacy rules.
- **The Right to Erasure (The "Wipe" Switch):** To comply with Section 19 of the 2019 Act (Right to Erasure), we implement a cryptographically secure, local SQL wiping routine. When an agent taps [Wipe My Data], the application executes a zero-fill overwrite of the local SQLite file and drops all active database keys, leaving the device completely clean.

2. Technical Engineering: Bypassing the SMS Permission Wall

Google's progressive security constraints on the `READ_SMS` permission (introduced in Android 10 and tightened in later versions) present a high risk of app rejection on the Google Play Store. To bypass this barrier, KioskControl uses a **Notification Listener Service**.

Instead of intercepting standard SMS protocols, KioskControl monitors incoming notifications generated by network dialers, system alerts, and push notifications.

2.1 Kotlin Android Notification Listener Implementation

```
package ug.kioskcontrol.services

import android.service.notification.NotificationListenerService
import android.service.notification.StatusBarNotification
import android.content.Intent
import android.util.Log
import java.util.regex.Pattern

class TransactionNotificationListener : NotificationListenerService() {

    override fun onNotificationPosted(sbn: StatusBarNotification?) {
        if (sbn == null) return

        val packageName = sbn.packageName
        val extras = sbn.notification.extras
        val title = extras.getString("android.title") ?: ""
        val text = extras.getCharSequence("android.text")?.toString() ?: ""

        // Target designated mobile money carriers and system notifications
        if (packageName.contains("mtn") || packageName.contains("airtel") ||
            packageName.contains("telecom") || title.contains("Money") ||
            title.contains("MoMo")) {

            parseNotificationPayload(text, packageName)
        }
    }

    private fun parseNotificationPayload(text: String, origin: String) {
        try {
            // Pattern Match for MTN Cash In / Deposit Notifications
            val mtnDepositPattern = Pattern.compile(
                "You have received UGX (\\d{1,3}(,\\d{3})*) from (\\d+). New ba
            )
```

```

        val matcher = mtnDepositPattern.matcher(text)

        if (matcher.find()) {
            val amountString = matcher.group(1).replace(",", "", "")
            val sender = matcher.group(3)
            val balanceString = matcher.group(4).replace(",", "", "")

            val amount = amountString.toDouble()
            val balance = balanceString.toDouble()

            broadcastTransaction(
                type = "Deposit",
                network = "MTN",
                amount = amount,
                counterparty = sender,
                parsedBalance = balance,
                rawMessage = text
            )
        }
    } catch (e: Exception) {
        Log.e("ParserError", "Failed parsing payment notification: ${e.message}")
    }
}

private fun broadcastTransaction(
    type: String,
    network: String,
    amount: Double,
    counterparty: String,
    parsedBalance: Double,
    rawMessage: String
) {
    val intent = Intent("ug.kioskcontrol.TRANSACTION_DETECTED")
    intent.putExtra("type", type)
    intent.putExtra("network", network)
    intent.putExtra("amount", amount)
    intent.putExtra("counterparty", counterparty)
    intent.putExtra("parsedBalance", parsedBalance)
    intent.putExtra("rawMessage", rawMessage)
    sendBroadcast(intent)
}
}

```

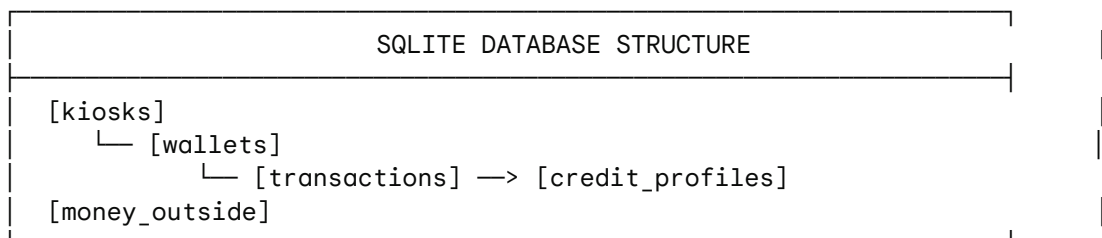
2.2 System Fallback Scheme: The Verification Check

If a network notification fails to register due to battery-saving policies or device restart states, the app falls back to a **"Reconciliation Request"** workflow:

1. At 6:00 PM, the system prompts the operator: *"No new notifications detected in the last 2 hours. Is your active MTN Float still UGX X,XXX?"*
2. The operator taps **[Verify Balance]**, opening a local numeric form to input the true balance.
3. The database updates, logs the manual override, and highlights the manually verified entry to the owner at closing.

3. Database Schema Design (The Credit Seed)

To support the Phase 3 credit scoring upgrade, the local schema is engineered from day one to log metadata that commercial banks (like Centenary Bank, Stanbic, and DFCU) require for small-business risk modeling.



```
-- Kiosks Configuration Model
CREATE TABLE IF NOT EXISTS kiosks (
  id TEXT PRIMARY KEY,
  name TEXT NOT NULL,
  district TEXT NOT NULL,
  owner_id TEXT NOT NULL,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- Wallet Resource Balances
CREATE TABLE IF NOT EXISTS wallets (
  id TEXT PRIMARY KEY,
  kiosk_id TEXT,
  carrier_type TEXT CHECK(carrier_type IN ('MTN', 'Airtel', 'Bank', 'PayWay'))
  current_float REAL DEFAULT 0.0,
  current_cash REAL DEFAULT 0.0,
  FOREIGN KEY(kiosk_id) REFERENCES kiosks(id)
);

-- Transaction Reconciliation Model (Automated Registry)
CREATE TABLE IF NOT EXISTS transactions (
  id TEXT PRIMARY KEY,
  wallet_id TEXT,
  type TEXT CHECK(type IN ('Deposit', 'Withdrawal', 'Expense', 'Commission'))
  amount REAL NOT NULL,
  commission_earned REAL DEFAULT 0.0,
  recorded_balance REAL NOT NULL,
  counterparty TEXT,
  timestamp TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  raw_payload TEXT,
  source_type TEXT CHECK(source_type IN ('Notification', 'SMS_Legacy', 'Manual_
  security_flag TEXT CHECK(security_flag IN ('Verified', 'Mismatch', 'Suspiciou
  FOREIGN KEY(wallet_id) REFERENCES wallets(id)
);

-- Money Outside Tracking (Informal Credit Logs)
CREATE TABLE IF NOT EXISTS money_outside (
  id TEXT PRIMARY KEY,
```

```
kiosk_id TEXT,
party_name TEXT NOT NULL,
transaction_direction TEXT CHECK(transaction_direction IN ('Lent_Out', 'Borro
principal_amount REAL NOT NULL,
repaid_amount REAL DEFAULT 0.0,
repayment_target_date DATE,
cleared_timestamp TIMESTAMP,
status TEXT CHECK(status IN ('Active', 'Settled', 'Overdue')) DEFAULT 'Active
FOREIGN KEY(kiosk_id) REFERENCES kiosks(id)
);

-- Credit Telemetry Engine (Anonymized Scoring Logs)
CREATE TABLE IF NOT EXISTS credit_profiles (
  id TEXT PRIMARY KEY,
  kiosk_id TEXT,
  rolling_90_day_volume REAL NOT NULL,
  daily_average_transactions INTEGER NOT NULL,
  max_single_drawdown REAL NOT NULL,
  float_stockout_count INTEGER DEFAULT 0,
  overdue_loans_count INTEGER DEFAULT 0,
  last_reconciled_at TIMESTAMP,
  kiosk_stability_score REAL DEFAULT 0.0,
  FOREIGN KEY(kiosk_id) REFERENCES kiosks(id)
);
```

3.1 The Kiosk Stability Score (KSS)

Using the database schema above, the app calculates a daily **Kiosk Stability Score (KSS)** locally on the device using five core factors:

$$KSS = w_1(V) + w_2(C) - w_3(S) - w_4(D) + w_5(R)$$

Where:

- *V* (Volume): Rolling 90-day transaction volume, proving revenue capacity.
- *C* (Consistency): Number of active trading days out of the last 90.
- *S* (Stockouts): Count of times float balances hit zero during peak trading hours.
- *D* (Defaults): Overdue loans marked in the **Money Outside** ledger.
- *R* (Reconciliation): Number of times the owner successfully reconciles cash balances with a perfect match at closing.

4. Monetization & The Competitive Moat Strategy

MONETIZATION STAGE PIPELINE	
Stage 1: Kiosk Subscriptions (UGX 30,000 / month)	
Stage 2: Bank Referral Commissions (1.5% - 2.0% on capital advances)	

4.1 Monetization Pipeline

- **The Baseline SaaS Model:** A multi-tier subscription model:
 - *Free Tier:* 1 Kiosk, basic transaction history parsing, offline tariff calculator.
 - *Pro Tier (UGX 30,000/month):* Multi-kiosk cloud dashboard integration, advanced employee anti-theft tracking, automated WhatsApp reconciliation reminders.
- **The Financial Infrastructure Play (The Scale Weapon):** Once we secure a critical density of 500+ active agents, we transition to our primary monetization engine: **B2B Bank Lending Referrals**.
 - Commercial banks like Centenary Bank and DFCU struggle to evaluate the credit risk of informal retail agents because cash velocity is invisible on paper records.
 - With the agent's consent, KioskControl exports their anonymized, high-fidelity **KSS Credit File** directly to partner bank credit divisions.
 - When the bank issues a UGX 5,000,000 working capital loan, KioskControl captures a **1.5% to 2% referral commission** on the disbursement. This transforms the business model from a small-scale software subscription into a central infrastructure platform for national liquidity lending.

4.2 The Competitive Moat (MTN & Airtel Defensibility)

Why can't MTN or Airtel simply build this feature themselves and kill our market?

1. **The Neutral Platform Advantage:** Over 80% of successful kiosks in Kampala operate as multi-network businesses. They handle MTN MoMo, Airtel Money, Agent Banking (Centenary/Stanbic), and utility portals (PayWay) simultaneously. MTN will never build a ledger that tracks and optimizes Airtel float, and Airtel will never build an app that reconciles Stanbic cash drawers. KioskControl's absolute neutrality is its primary competitive shield.
2. **The Local Association Integration:** Our exclusive pilot rollout via the Vice President of the Mobile Money Agents Association embeds KioskControl directly into the institutional fabric of the market. This creates high switching costs and established, peer-to-peer first-mover trust that no corporate telecom can replicate from above.
3. **The Proprietary Credit Asset:** Once the platform secures 90 days of structured